

Backend-focused Software Engineer with 3+ years of experience building cloud, IoT, EV charging, and platform systems. Strong hands-on expertise in **Go/Golang**, **OCPP 1.6J**, AWS, Google Cloud Pub/Sub, MQTT, PostgreSQL/TimescaleDB, Docker, Terraform, event-driven architecture, and automated integration testing. Experienced in owning distributed backend services from architecture and API design through implementation, infrastructure, observability, production debugging, and real charger validation.

EXPERIENCE

- **EVRA Solutions** Remote
Backend Engineer Apr 2026 – Present
 - **Go charging-session service:** Built and evolved EVRA's standalone Charging Session Service, establishing production-ready configuration, structured logging, health/readiness checks, PostgreSQL and TimescaleDB connectivity, migrations, containerization, and automated tests.
 - **OCPP transaction lifecycle and telemetry:** Implemented **StartTransaction**, **MeterValues**, and **StopTransaction** workflows with validation, charger/connector resolution, idempotent session creation, transaction-based lookup, session closure, and billing-grade energy and SoC aggregation.
 - **Event-driven architecture migration:** Migrated charger and charging-session workflows from direct MQTT consumption to normalized **Google Cloud Pub/Sub** per-action topics; added emulator-backed integration tests for envelope parsing, deduplication, routing, and response publishing.
 - **Authorization, multi-tenancy, and smart charging:** Integrated RFID/driver authorization and concurrent-session limits; enriched sessions with company, site, driver, vehicle, and energy data; and implemented site power aggregation, reserved-power tracking, debounce logic, and Vortex-triggered profile recalculation.
 - **Production-readiness and hardware validation:** Built contract, event-sourcing, OCPP lifecycle, and end-to-end test suites across edge-device, charger, user, and charging-session services, and validated complete flows against physical **Monta OCPP 1.6J** charger hardware.
- **NeoCharge** Remote
Senior Software Engineer Jan 2025 – Apr 2026
 - **EV charging backend and OCPP orchestration:** Designed and maintained modular Go services for charger onboarding, session lifecycle, telemetry, Remote Start/Stop, charging profiles, and smart scheduling across **Autel** and **Wallbox** chargers.
 - **Secure IoT device onboarding:** Designed certificate-based provisioning and re-enrollment flows using device-generated CSRs, X.509 certificates, mTLS, AWS IoT Core policies, and exclusive thing-principal associations while keeping private keys on-device.
 - **Messaging and telemetry platforms:** Built a custom **Mochi-MQTT** communication layer with Redis-backed persistent sessions and developed event-driven telemetry pipelines using AWS IoT Core, Kinesis, S3, DynamoDB, Amazon MQ, and Lambda.
 - **Cloud infrastructure and delivery:** Migrated workloads toward AWS ECS/Fargate, provisioned infrastructure with Terraform, and implemented GitHub Actions pipelines for testing, Docker image publishing, and deployment; reduced AWS costs by approximately **20%**.
 - **Integrations, observability, and production support:** Integrated Enode, LEAP, RevenueCat, EVOX, OpenEI, WattTime, and energy-platform APIs; added Datadog monitoring and resolved production issues across chargers, messaging, TLS, containers, and cloud permissions.
- **Octopus Digital (Avanceon)** Lahore, Pakistan
Full Stack Developer Apr 2023 – Dec 2024
 - **Industrial IoT platform development:** Delivered end-to-end **.NET Core**, **SQL**, and **Angular** features for OmniConnect, including asset/tag hierarchies, configuration workflows, paginated APIs, alarms, and real-time industrial data views.
 - **Bulk configuration automation:** Optimized Excel/CSV import workflows for thousands of assets and tags, reducing configuration time by **98%** from hours to minutes.
 - **Security, auditability, and operations:** Implemented audit trails, time-based access controls, user blocking with history retention, advanced filtering, and operational dashboards for industrial and fuel-management products.
 - **AI-assisted platform capabilities:** Built POCs with Azure OpenAI, OpenAI Assistants API, and Semantic Kernel; one evolved into OmniGPT, a context-aware, tool-augmented AI capability within the platform.

EDUCATION

- **FAST NUCES** Lahore, Pakistan
Bachelor of Science in Computer Science *2019 – 2023*

CLIMATE TECH APPRENTICE

- **C2C LaunchPad** Remote
Climate Tech Apprentice
 - **Climate and market research:** Explored seven climate-tech sectors and 93 Project Drawdown solutions, researching business models, innovation gaps, electrification, energy infrastructure, and EV charging.
 - **Cross-functional communication:** Strengthened climate-market analysis, stakeholder engagement, and the ability to communicate technical solutions in a commercial and sustainability context.

TECHNICAL SKILLS

- **Languages & Backend:** Go/Golang, Python, FastAPI, .NET Core, REST APIs, webhooks, GORM, Fiber, Echo, pgx, Domain-Driven Design, Clean Architecture, unit, integration, contract, and end-to-end testing
- **EV Charging, IoT & Messaging:** OCPP 1.6J, MQTT, Mochi-MQTT, Mosquitto, Google Cloud Pub/Sub, AWS IoT Core, mTLS/X.509, charger commands, smart charging, demand response, telemetry ingestion, event-driven architecture
- **Databases:** PostgreSQL, TimescaleDB, SQL Server, DynamoDB, Redis, Amazon RDS, Aurora PostgreSQL, high-ingestion telemetry modeling, schema design, query optimization
- **Cloud, DevOps & Observability:** AWS ECS/Fargate, Elastic Beanstalk, Lambda, S3, Kinesis, EventBridge, ECR, Amazon MQ, Terraform, Docker, Docker Compose, GitHub Actions, LocalStack, testcontainers-go, Datadog, CloudWatch, Grafana
- **Frontend & AI:** TypeScript, JavaScript, Angular, HTML/CSS, Azure OpenAI, OpenAI Assistants API, Semantic Kernel